

Cellula Week 6

Project 3.1 — Evaluator-Generator Workflow

Team Members

Member	Name	Role
Member 1	Alaa AbdelHakeem	Data Ingestion & UI
Member 2	Zeina Mostafa	Core LLM Engineering
Member 3	Alaa Khamees	Orchestration, Caching & Feedback Loop

This report summarizes the work distribution for Project 3 (Evaluator-Generator Workflow), covering the team division and the implementation tasks handled by each member in Part 1: Data Ingestion & UI.

Member 1 — Alaa AbdelHakeem

Data Ingestion & User Interface (The Frontend & Loader)

- Built the Streamlit interface.
- Handled file uploads (PDF, DOCX, TXT, PPT, Code, WAV) and URLs.
- Implemented Speech-to-Text for WAV files.
- Processed data: chunking, generating embeddings, and storing them in a Vector DB.

Member 2 — Zeina Mostafa

Core LLM Engineering (Generator & Evaluator Prompts)

- Designed the Generator LLM to strictly answer from the retrieved context without hallucinating.
- Designed the Evaluator LLM to grade the answer based on accuracy, relevance, and completeness.
- Managed the isolated, independent memories for both models.

Member 3 — Alaa Khamees

Orchestration, Caching & Feedback Loop (The System Architecture)

- Used LangChain Expression Language (LCEL) to connect the Generator and Evaluator.
- Built the 4-iteration feedback loop logic.
- Integrated Redis caching to save embeddings and previous answers.
- Handled logging, error management, and overall system integration.